CAPA

Table of Contents

Abstract

This project aims to assist Cars 4 You, an online resale company that sells cars, in developing an efficient machine learning algorithm capable of evaluating the price of a car based on the user's provided information.

1. carId was set as the index. then brand , model, transmission and fueltype columns was normalized and sorted into bins, as there were many entries with typos, lowered and upper case variations.
2. All entries after the year 2020 was removed , as the database is from 2020, so it doesn't make sense having cars from after that time frame, and then we remarked that all cars before 1980 had wrong so we elected to just drop entries.
3. All negative millage , tax and mpg were assigned to nan values , and then dropped them , while rearranging data columns to have price on the last column
4. All engine sizes below 0.6 to nan values and then dropped them.
5. All paint quality values above 100 and bellow 0 were assingned to nan and dropped them.
6. All previous owner with less than 0 owners were assinged to np.nan and dropped them, while also rounding all the previous owners to the nearest integer.
7. The damage collum was dropped since it contained no values other than 0.
8. All categorical cols with nan values were filled with the string "Unknown"
9. The target variable, price was separated the rest of the other variables.
10. We split our data into 2 sets, train and validation
11. Float columns with nan values was filled with the mean values of the collum and the int columns with median value
12. Outliers by using winsorization and log transformation.
13. The model was scaled using minmaxscaler
14. Correlation was tested using spearman method and our regression using mi-scores
15. Significance was tested by , using chi squared for the numerical features and for categorical features we user cramer's test
16. Independence between categorical features was tested by first enconding the categorical features LabelEncoder() and then by measuring with mutual_info_classif function.
17. ANOVA was calculated to test significance of categorical types.
18. RFE was used to the top features.
19. Multicollinearity was assessed , and none was found.
20. Lasso was used, and all other variables pass the lasso test.
21. One hot enconding to transform categorical variables into a format that can be read by the model.
22. Kfold cross validation was used and an R^2 and RMSE score were obtained.
23. A random forest regressor was also testeded and found to be overfitted